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A Dual Deep Neural Network Approach for Discriminating Internal Faults and Inrush Currents in Transformers Under CT Saturation

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ABSTRACT The dependability and security of transformer differential protection can be compromised by current transformer (CT) saturation, which distorts current measurements and may lead to the maloperation of protective relays. This paper proposes a dual deep neural network (dual-DNN) approach to accurately distinguish internal faults from simultaneous inrush currents, even under CT saturation. The proposed architecture consists of two stacked denoising autoencoder (SDAE)-based networks: a compensation DNN (Com-DNN) and a discrimination DNN (Dis-DNN). The Com-DNN mitigates the adverse effects of CT saturation by learning to recognize and correct distortions in the current signals, while the Dis-DNN discriminates between inrush currents and internal faults. The dual-DNN approach was evaluated using a 23-kV power transformer under various scenarios, including simultaneous inrush and fault events. Simulation results demonstrate that the proposed approach delivers robust performance and stability under challenging conditions. A comparative analysis with a conventional artificial neural network (ANN) and three other existing methods confirms that the dual-DNN approach offers superior dependability and security. Furthermore, hardware implementation validates its feasibility and effectiveness for real-world deployment.

INDEX TERMS Current transformer saturation, differential protection, dual deep neural network, inrush current, internal fault, power transformer, stacked denoising autoencoder.

I. INTRODUCTION

A. Motivation and Incitement

Power transformers can experience various transient conditions that may saturate the cores of current transformers (CTs) used for differential protection. CT saturation during external faults can lead to unwanted operations, resulting in nuisance trips of circuit breakers. Phasor estimation is also significantly compromised, which in turn affects the performance of overcurrent, distance, and differential protection [1]. Distinguishing inrush currents from internal faults is particularly challenging due to the risk of nuisance tripping caused by magnetizing inrush. Additionally, the security of transformer differential protection is threatened during CT saturation, as nuisance trips in response to external faults.

Traditionally, malfunctions under CT saturation are mitigated using harmonic blocking schemes or dual-slope characteristics in differential protection. However, these methods can compromise dependability and reduce the effectiveness of differential protection when internal faults occur during CT saturation [2].

In contrast, deep learning–based approaches provide significant advantages by capturing nonlinear relationships and enhancing both the dependability and security of protection schemes under diverse and uncertain operating conditions. Therefore, they are regarded as promising solutions for reliably discriminating internal faults from inrush currents in transformers, even in the presence of CT saturation.

B. Literature Review

To prevent erroneous differential protection during transformer energization, harmonic-based solutions, such as harmonic blocking and harmonic restraint techniques, have been widely recommended [3], [4]. Although these methods generally provide high dependability, they can compromise security when the harmonic content of inrush current is reduced. Additionally, the harmonic restraint technique may result in undesired blocking when energizing a faulty transformer, particularly if the healthy phases contain high harmonic content [5]. While the harmonic blocking scheme offers better security, it can become unstable when a faulty transformer is energized. Moreover, power transformers with modern cores may produce low levels of second harmonic content during energization, leading to the misoperation of harmonic-based methods [6]. Furthermore, the second harmonic content may increase on long transmission lines as shunt reactors or series capacitors are added during system expansion [5]. These challenges highlight the need for transformer differential protection schemes that can accurately distinguish internal faults from inrush currents, thereby enhancing security during energization and in scenarios where internal faults coincide with inrush events.

The literature describes various techniques for discriminating between inrush currents and internal faults, which can be broadly classified into three categories. The first category employs signal-processing techniques, such as the wavelet transform (WT) and the Fourier transform (FT), to extract spectral and waveform features (detection points) [6]–[9]. However, these methods have notable limitations, including susceptibility to noise and a heavy reliance on predefined thresholds, which can compromise the dependability and security of transformer differential protection.

The second category involves model-based techniques that utilize estimation methods, such as the Kalman filter [10], to distinguish internal faults from inrush currents with greater accuracy. The primary limitation of this approach is its inability to precisely model the complex physical behavior of power transformers. To address the sparsity issue encountered in traditional support vector machines (SVMs), a least-squares (LS) SVM is employed to classify inrush currents and internal faults in large-feature problems [11]. Dwelling time-based methods have also been introduced to overcome the drawbacks of harmonic-based approaches. Other approaches, such as derivation and zero-band techniques, are capable of detecting inrush currents [12]–[14]. However, their sensitivity to noise and other disturbances can result in false detections [14]. Additionally, a detection index based on the dead angle [15] has been proposed to identify inrush currents. However, variations in waveform distortion width may lead to misjudgments.

The final category includes intelligent methods that can accurately differentiate internal faults from inrush currents without relying on harmonic information or assumptions about the optimal mathematical representation of power transformers [16]–[18]. Artificial neural networks (ANNs), machine learning algorithms, and hybrid methods have been used to discriminate between these two abnormalities. Approaches based on correlation analysis [16], and SVMs [17] have yielded promising results. However, these methods typically lack the ability to independently extract time-domain features and often require pre-processing techniques such as WT to extract relevant features before detection. Their performance may degrade when dealing with complex feature patterns or external faults. Additionally, they generally do not consider CT saturation during either internal or external faults. To overcome these limitations, a convolutional neural network (CNN) combined with the XGBoost technique has been proposed to identify abnormal conditions in power transformers [18].

As deep learning has recently emerged as a powerful tool for automatic feature extraction, DNNs have been applied to various power system applications, including fault recognition and signal estimation [19]–[21]. A deep-learning-based transformer differential protection scheme was also developed to distinguish internal faults from inrush currents [5]. Unlike conventional methods, this approach does not rely on harmonic information or predefined thresholds but instead utilizes sampling data and waveform patterns. However, it remains vulnerable to CT saturation during external faults. In [22], source and target convolutional neural networks (SCNN and TCNN) were employed for transformer fault detection by pre-training on unsaturated segments and suppressing saturated features. The training process of the SCNN–TCNN framework is relatively complex, as the TCNN is heavily dependent on the SCNN for accurate classification. Consequently, classification accuracy may be compromised if the features extracted by the SCNN fail to generate reliable feature labels for the TCNN.

C. Contribution and Organization of the paper

Motivated by the limitations identified in [5], this paper presents an improved differential protection scheme for power transformers, referred to as the dual-DNN approach. This new approach discriminates between internal faults and inrush currents while also addressing the challenges posed by CT saturation during external faults. First, a compensation DNN (Com-DNN) is employed to mitigate the adverse effects of CT saturation, using stacked denoising autoencoders (SDAEs). Once CT saturation is compensated for, a discrimination DNN (Dis-DNN) is used to distinguish inrush currents from internal faults. The Dis-DNN leverages unsupervised feature learning through SDAEs, followed by supervised fine-tuning. This dual-DNN approach effectively handles simultaneous internal faults and inrush currents, achieving rapid detection within less than one cycle. The main contributions of this paper are summarized as follows:

1) The dual-DNN approach mitigates the adverse effects of CT saturation through the Com-DNN and accurately discriminates internal faults from inrush

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currents by extracting relevant features and patterns using the Dis-DNN.

2) The Dis-DNN minimizes the need for predefined thresholds by employing supervised learning trained with time-aligned event labeling.

The remainder of this paper is organized as follows. Section II presents the proposed dual-DNN approach, including the Com-DNN and the Dis-DNN. Section III validates the effectiveness of the approach using PSCAD/EMTDC-simulated waveforms. Section IV describes the hardware implementation on a DSP-based prototype. Finally, Section V provides concluding remarks.

II. Dual Deep Neural Network Approach

This section presents the essential features of comprehensive differential protection of power transformers. First, the distorted signals measured from both sides of all CTs should be compensated to avoid erroneous operation during both external faults and internal faults. Second, normal and abnormal conditions (such as inrush currents or internal faults) are detected based on the differential current after compensation step; no specific threshold is required. Finally, internal faults are distinguished from other disturbances using the proposed dual-DNN approach. Hence, the tactful combination of the Com-DNN and Dis-DNN is implemented in sequence as shown in Figure 1. It begins with data acquisition and data preprocessing from system modelling in PSCAD/EMTDC, as shown in the first subfigure of Figure 1. In the second subfigure, The Com-DNN utilizes the secondary currents of CT1 and CT2 to precisely estimate their primary currents (without saturation), aiming to reduce the negative impact of CT saturation. Once the Com-DNN is applied, the data is prepared for the Dis-DNN, which differentiates between internal faults and inrush currents, as illustrated in the fourth subfigure of Figure 1.

A. Data Preprocessing for the Com-DNN

Data Window (DW) is a technique applied in power system protection for fault detection, direction estimation, time-series forecasting, and fault classification. It yields promising results at moments when there is significant fluctuation in the waveform. The results based on the data window become noticeable when dealing with abnormal conditions. Inspired by this concept, we used a DW originally proposed in [23] to detect power swings in a transmission line. Considering the measured current from a CT, denoted as $i_{CT,input} = \{x_1, x_2, \ldots, x_n\}$, a set of DWs of the measured current is expressed in (1) as follows:

$$\text{Set of DWs} = \begin{bmatrix} x_1 & x_2 & \cdots & x_{m-1} & x_m \\ x_2 & x_3 & \cdots & x_m & x_{m+1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n-m+1} & x_{n-m+2} & \cdots & x_{n-1} & x_n \end{bmatrix} \quad (1)$$

DNN-based methods are well-known for producing inconsistent results when there is variation in the magnitude of datasets. To address this issue, it is essential to normalize the input datasets for the DNN. By applying normalization techniques to scale the data into a normalized range of [-1,1], error resulting from variation in magnitude can be minimized. As a result, training is accelerated [24]. One row from (1) is taken as a reference for normalization:

$$i_{Norm} = \frac{i_k}{\max\limits_{1 \le k \le N} i_k}, \; k = 1, 2, \ldots, N \quad (2)$$

where i_{Norm} denotes the normalized value of the input dataset and N corresponds to number of samples in one cycle, which is 64 for the window size.

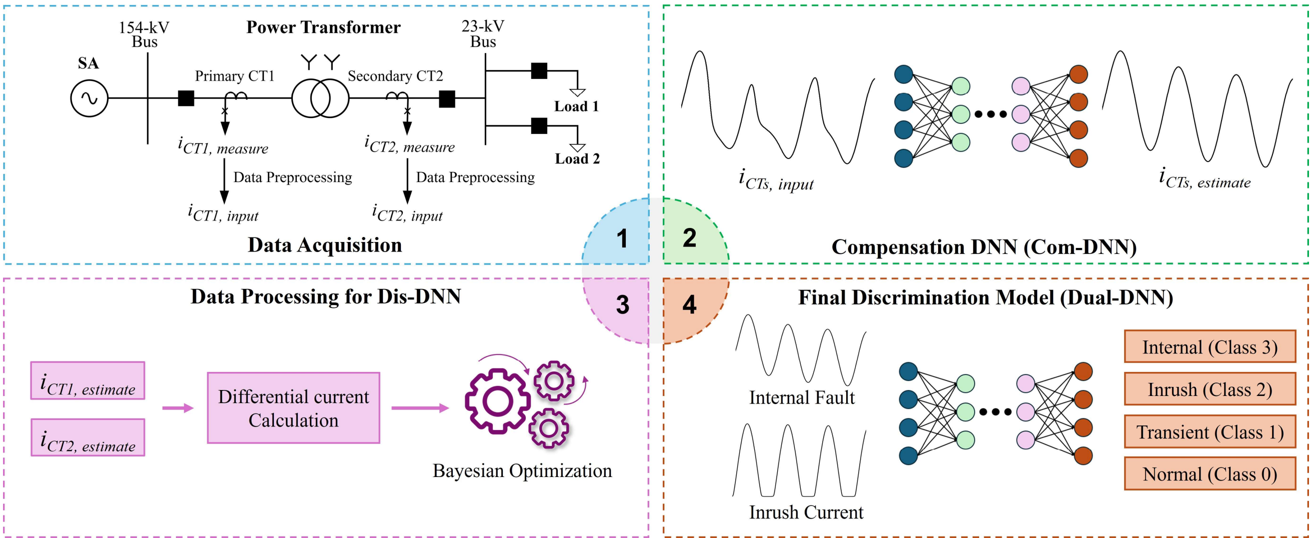

FIGURE 1. Proposed framework for discrimination between internal faults and inrush currents based on the dual-DNN model

B. CT Saturation Compensation Based on the Com-DNN

Large fault magnitude, primary and secondary time constants, and burden can heavily influence CT saturation. Although saturated inrush current is possible, it is rare and therefore not considered in this study. CT saturation causes a considerable difference between the desired and measured values of the secondary current, consequently producing malfunctions in transformer differential protection due to a false differential current. The LS-based method used in [25] mitigates the adverse effects of CT saturation and DC offset; however, the physical representation of a CT may differ from its actual behavior, potentially leading to incorrect estimation of the undistorted current. Therefore, we apply the Com-DNN model prior to discrimination, as shown in the second subfigure of Figure 1. The Com-DNN automatically extracts features from the distorted secondary current and compensates for the distortion in the signal [22]. By generalizing various parameters of CT saturation and noises, the Com-DNN ensures that the Dis-DNN operates efficiently and remains robust against CT saturation.

C. Characteristics of Inrush Currents and Internal Faults on the Data Window

The waveform in the DW, as described in (1), contains many distinctive characteristics at each sample point. The results from the Com-DNN are immediately used to calculate the differential current for discriminating between inrush currents and internal faults. The Dis-DNN captures unique features that distinguish internal faults from inrush currents. The Dis-DNN performs multiclass pattern classification with four conditions: normal, transient, inrush current, and internal fault. Figure 2 illustrates the instantaneous differential current, where the DW is applied with a window length of one cycle.

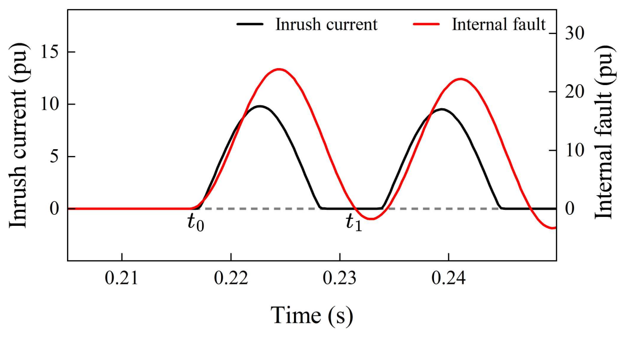

FIGURE 2. **Differential current when an inrush current (black) and an internal fault (red) happen**

Before time t_0, both the inrush current and the internal fault current are approximately zero. This indicates that the transformer is operating under normal conditions, with no abnormal differential current present. In this case, no disturbance is detected. At time t_0, both currents suddenly increase due to either an inrush or an internal fault event. During this period, it is difficult to distinguish the source of abnormality, as both conditions result in large and similar spikes in the differential current. Therefore, no protective action is taken immediately, and the system waits for additional samples to observe further behavior. Similarly, when a faulty transformer is energized, it exhibits a large differential current. As shown in Figure 2, the value of differential current becomes positive after t_0, indicating the occurrence of abnormal conditions, which can be referred to transient state at point t_0. However, an internal fault and an inrush current exhibit distinctive characteristic after point t_1. In the case of an internal fault, the value becomes zero or negative for fault inception angles of 0° and 90°. When such behavior is detected, the Dis-DNN promptly transitions to identifying an internal fault; otherwise, it identifies an inrush current. The sampling delay between points t_0 and t_1 is less than one cycle, specifically, 58 samples when one cycle corresponds to 64 samples. This short delay allows clear discrimination between an internal fault and an inrush current.

To enable the Dis-DNN to effectively distinguish and visualize each condition, one-hot encoding was applied for labeling during training, as shown in TABLE I. This encoding allows the Dis-DNN to accurately learn the distinct features of each condition. After training, the output is converted to numerical classes to clearly discriminate among the conditions during testing and implementation.

TABLE I.
Labeling with one-hot encoding for the Dis-DNN

Conditions	One-hot encoding	Class
Normal	0 0 0 1	0
Transient	0 0 1 0	1
Inrush Current	0 1 0 0	2
Internal Fault	1 0 0 0	3

D. Unsupervised Feature Learning and Supervised Fine Tuning in the Dis-DNN

SDAEs are widely used for data prediction and reconstruction and have demonstrated significant potential in representing electrical measurements and protection. The choice of implementing SDAEs in this study is related to their ability to generalize waveform models in power systems, where disturbances such as random noise and high-frequency components caused by external factors are present. Therefore, SDAEs can effectively mitigate these interferences by capturing the essential features of the input data while filtering out the aforementioned disturbances.

1) Unsupervised Feature Learning

Unsupervised feature learning can be easily achieved by adopting an autoencoder (AE), which is effective at extracting features from nonlinear input datasets. In this paper, we use SDAEs, as introduced in [26], to achieve feature learning that is insensitive to the random noise encountered in real-world applications. Typical noise interference in power systems may contain an unspecified frequency spectrum, which can be approximated by

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additive white Gaussian noise and mathematically expressed as follows.

$$SNR_{dB} = 10 \cdot log(A_{signal}/A_{noise}) \qquad (3)$$

The architecture of SDAEs is similar to that of SAEs, and the training process ensures that the output vector from the reconstruction layer closely approximates the input vector, free of random noise introduced during measurement, for each unlabeled data sample $x \in R^n$ in the training dataset. Thus, the training process is considered complete when the hidden layers yield effective feature representations of the training dataset. Given a noisy input vector $\tilde{x}$, the output vector of the SDAEs, $h(x)$, is computed as:

$$h(\tilde{x})=W_{de} \cdot f\left(W_{en} \cdot \tilde{x}+b_{en}\right)+b_{de} \qquad (4)$$

where $f(z)$ is the nonlinear leaky Rectified Linear Unit (Leaky ReLU) activation function. W_{en} is the $n_h \times n$ weight matrix linking the input and hidden layers, and b_{en} is the bias vector for the hidden layer. W_{de} is the $n \times n_h$ weight matrix linking the hidden and reconstruction layers, and b_{de} is the bias vector for the reconstruction layer. The goal is to minimize the difference between the clean input vector x and $h(\tilde{x})$; therefore, a cost function is required to measure performance across the entire dataset. Specifically, the cost function J includes a root mean square error term, a weight decay term, and a sparsity penalty term.

$$J = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\|h(x^i)-x^i\|^2} + \lambda\sum_{i=1}^{2}\|w_i\|^2 + \beta\sum_{j=1}^{n_h} KL\left(\rho\middle|\tilde{\rho}_j\right) \qquad (5)$$

The first term measures the error between the input and output datasets. The second term constrains the range of the weight distribution to prevent the autoencoder from overfitting. The third term controls the weight updates within the cost function. $KL(\rho|\tilde{\rho}_j)$ is a sparse constraint that measures the similarity between the desired and actual distributions of the weights and bias initializations. The use of a small sparsity term ρ ensures that the training parameter distribution remains close to zero; therefore, it prevents parameter explosion and vanishing gradients during updates. The calculation is presented below:

$$KL\left(\rho\middle|\tilde{\rho}_j\right) = \rho \; log\left(\frac{\rho}{\tilde{\rho}_j}\right) + (1-\rho) \; log\left(\frac{1-\rho}{1-\tilde{\rho}_j}\right) \qquad (6)$$

The inclusion of the sparsity term in the cost function ensures that relevant feature representations of the input vector x are obtained. Using the backpropagation algorithm [27], we iteratively update the training parameters based on the cost function J. After several epochs, J is expected to converge with the minimum error, at which point the unsupervised feature learning by the DAEs is considered complete.

Figure 3 presents the unsupervised feature learning flow. We use the DW of (1) to allow the DAEs to capture fluctuations in the input datasets; thus, the DAEs can accurately generate estimated outputs at each instant of the data window. The input block in Figure 3 contains both internal faults and inrush currents, and this input block is used to train the DAEs to estimate the target output.

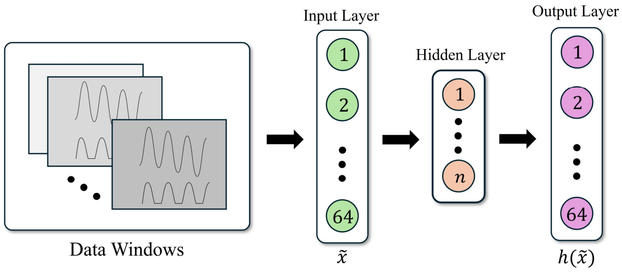

FIGURE 3. Unsupervised learning by DAEs; DWs are moved to the input layer

2) Supervised Fine Turning

Basically, the final model output separates the four classes defined in Section II: normal, transient, inrush current, and internal fault. As described above, a transient state is detected when minor fluctuations appear in the differential current. The unsupervised learning process is structured as a stack of hidden layers, where the parameters of the previous layers are frozen. The first hidden layer is established through the first DAE, and the feature representation from this first DAE serves as the input to the second DAE. After several similar iterations, the Dis-DNN consists of multiple stacked hidden layers, with the training parameters saved and later restored for the fine-tuning stage. At the final stage, an output layer is added to the Dis-DNN, initializing a new set of training parameters between the last hidden layer and the output layer. The weights w_i and bias values b_i of all DAEs are jointly trained using backpropagation; this structure is referred to as stacked denoising autoencoders (SDAEs). To discriminate internal faults from inrush currents, a SoftMax classifier is added to the output layer, enabling the description of each output as a probability; the calculation is expressed as:

$$h_\theta(x^i)=\begin{bmatrix} p(y^i=1|x^i,\theta) \\ p(y^i=2|x^i,\theta) \\ \vdots \\ p(y^i=k|x^i,\theta) \end{bmatrix} = \frac{1}{\sum_{l=1}^{k} e^{\theta_l^T x^i}} \begin{bmatrix} e^{\theta_1^T x^i} \\ e^{\theta_2^T x^i} \\ \vdots \\ e^{\theta_k^T x^i} \end{bmatrix} \qquad (7)$$

where y^i is the stochastic variable representing the output class corresponding to the input dataset x^i, and j denotes the output class. $\theta = (\theta_1^T, \theta_2^T, \dots, \theta_k^T)$ is the model parameter set. Consequently, the output of the SoftMax classifier is a 4-dimensional vector representing the four possible classes. The maximum probability for each class is determined as follows:

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$$Class(x^i) = argmax_{j=1,\dots,k}\, p(y^i{=}j|x^i,\theta) \tag{8}$$

Similarly, the SoftMax classifier is trained using the training dataset, iteratively optimized by the following cost function:

$$L= -\frac{1}{N}\left[\sum_{i=1}^{m}\sum_{j=1}^{k} S\{y^i{=}j\}log\frac{e^{\theta_j^T x^i}}{\sum_{l=1}^{k} e^{\theta_l^T x^i}}\right]+\frac{\sigma}{2}\sum_{i=1}^{k}\sum_{j=1}^{n}\theta_{ij}^2 \tag{9}$$

Where y^i is the i^{th} scalar value from the SoftMax output of (7), and S is the indicator function. σ is included in the cost function to penalize large parameter values, and L is strictly convex. Figure 4 illustrates the overall procedure by which the SDAEs distinguish an internal fault from an inrush current.

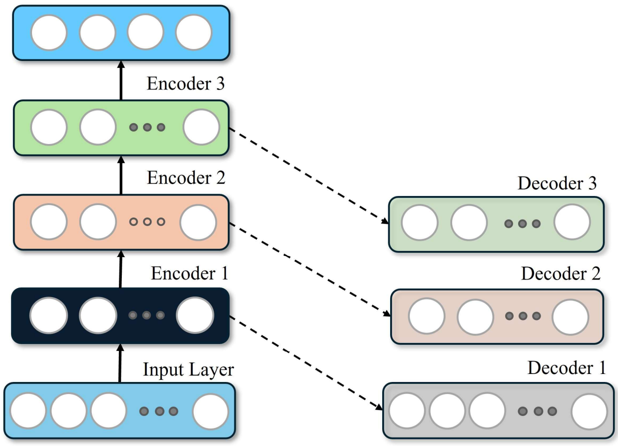

FIGURE 4. The staking process of SDAEs for the proposed Dis-DNN

E. Hyperparameters Tuning Using Bayesian Optimization

Training hyperparameters influence the overall performance of a DNN model. While the parameters can be selected through trial-and-error and empirical methods, various optimization strategies are also available to facilitate the search for optimized hyperparameters [28]. Each approach has its disadvantages: grid search may struggle with the dimensionality of large search spaces, while random search often yields results where individual hyperparameters are not easily interpretable.

Bayesian optimization (BO) is a sequential model-based approach used for hyperparameter tuning in machine learning and deep learning [29]. It builds a surrogate model to approximate the objective function $f(x)$ and uses an acquisition function to select the next configuration set x_t for evaluation. Based on the evaluated performance of $f(x_t)$, BO updates the surrogate model and repeats this process iteratively until a stopping criterion is met. The objective function $f(x)$ is the SDAE model, and BO is used to identify the best combination of hyperparameters in the search domain, represented as a 3-dimensional vector (learning rate, batch size, and number of neurons).

A Gaussian process is used as the prior belief, which is then distributed to model the SDAE function across each set of hyperparameters in the search space; this is referred to as the surrogate model. As the surrogate model iterates over specific hyperparameter sets, it produces probabilistic estimations of the objective function. The acquisition function, specifically the expected improvement (EI), evaluates these probabilistic estimations and selects x_t to optimize the search space. The Gaussian process is then updated with a new prior belief, recording the evaluation score, and the EI function determines x_t to be assessed. After multiple iterations, the final set of training hyperparameters is obtained when no further improvements are observed in the search space. The EI used to choose x_t is expressed as:

$$x_t = arg\, max_{x \in X}\, EI(x) \tag{10}$$

where

$$EI(x)=E\big(max\big(f(x)\text{-} f^*,0\big)\big) \tag{11}$$

and f^* is the maximum value that $f(x)$ has achieved during the optimization process.

III. Performance Evaluation

A. System Modeling and Data Acquisition

Figure 5 illustrates the typical three-phase 154/23-kV distribution system under study.

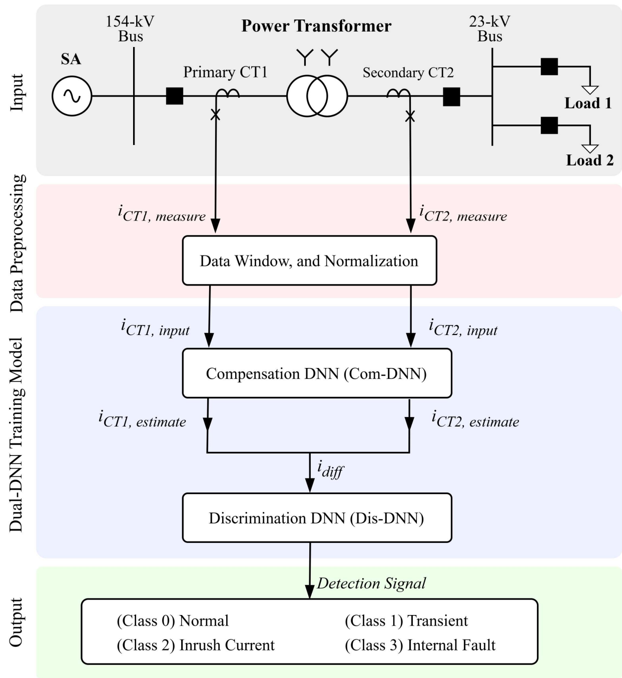

FIGURE 5. Single-line diagram of the PSCAD model and flowchart of the sequential steps in the dual-DNN approach

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The sampling frequency was set to 3840 Hz, which corresponds to 64 samples per cycle for a 60-Hz system, considering hardware implementation constraints. The simulation used a 40 MVA power transformer with a voltage ratio of 154/23 kV and a Y-Y configuration, with a positive leakage reactance of 10.99%. The source had positive- and negative-sequence impedances Z_1, Z_2 of (0.0419 + j0.8921) Ω/km, and a zero-sequence impedance Z_0 of (0.0128 + j0.0042) Ω/km. The turn ratios of CT1 and CT2 were 1200:5 and 2000:5, respectively. Note that the CT in Figure 5 refers to the JA built-in model in PSCAD, which accurately simulates the magnetization curve and provides a good representation of hysteresis loop [30].

We generated a dataset containing inrush currents and internal faults by varying the system parameters listed in TABLE II. The residual flux was considered in the simulations of the inrush current, as the characteristics and magnitude of the inrush current depend on the residual flux. The switching angle also significantly influences inrush current behavior. We shuffled all inrush current parameters, yielding 412,794 training datasets corresponding to 323 cases of inrush current patterns. After shuffling, the number of internal fault datasets was 218,538, corresponding to 171 cases of A-G faults.

TABLE II.
System parameters for the PSCAD/EMTDC simulation

	Inrush Current		Internal Fault	
	Switching angle (°)	Residual flux (%)	Fault Inception angle (°)	Winding location (%)
Minimum	0	-80	0	0
Maximum	180	80	180	80
Step	10	10	10	10

B. Network Construction and Training

TensorFlow is a mainstream deep learning platform developed by Google that streamlines the architecture and training of network models. We utilized it to construct and train both the saturation and discrimination models. The SDAE architecture was implemented using TensorFlow, enabling efficient training. To mitigate the vanishing gradient problem and enhance both accuracy and performance, the Leaky ReLU activation function was applied to the hidden layers, while the SoftMax activation function was used in the output layer. Consequently, the normal condition, transient, inrush current, and internal fault are determined in the last fully connected layer based on the probability estimation of the SoftMax function.

Prior to training, the input datasets were normalized within the range of [-1,1] using (2) above. We employed the Adam optimizer for gradient backpropagation to update the network weights and biases. A descending learning rate was used to optimize convergence; the rate was initially set to $1e^{-3}$ and was exponentially decayed after a certain number of epochs, which facilitated the learning process and helped prevent overfitting.

C. Performance Evaluation Indices

To address potential data imbalance in the dataset [33], we evaluated the performance of the Dis-DNN model using a confusion matrix, which provides three evaluation indices to validate the results as percentages:

$$ACC = \frac{TP+TN}{TP+TN+FP+FN} \quad (12)$$

$$DEP = \frac{TP}{TP+FN} \quad (13)$$

$$SEC = \frac{TN}{TN+FP} \quad (14)$$

where TP is the number of detected internal faults, and TN is the number of no-fault cases. FP, and FN refer to the numbers of normal conditions misclassified as faults that activate the differential protection, and faults misclassified as normal conditions that block the differential protection, respectively.

The accuracy (ACC) in (12) measures the ratio of correct predictions of faults and no-fault cases over all detections. Dependability (DEP), as defined in (13), reflects the ability to detect internal faults in the test datasets without undesired blocking of differential protection. It evaluates the decision-making of the differential protection to operate when internal faults occur during inrush conditions. Security (SEC) in (14) evaluates whether the Dis-DNN distinguishes the normal condition from internal faults without triggering differential protection. In other words, dependability measures the proportion of correctly identified positive cases, ensuring that minority classes are not overlooked, while security measures the proportion of correctly identified negative cases, preventing overprediction of the majority class. A balanced performance across both metrics provides strong evidence that the model can effectively handle class imbalance. To validate these indices, a comparative analysis is carried out at the 58th (N-6) and 61st (N-3) samples from an event, considering that the protection scheme typically operates after one cycle.

D. Simulation Results

To demonstrate the effectiveness of the dual-DNN approach, we compared it with the unidirectional and quartering-based similarity index from [31], the conventional second harmonic blocking scheme from [32], and the extended Kalman Filter (EKF) method from [10]. The graphical illustrations and evaluation indices make it abundantly clear that the dual-DNN approach is highly effective under both inrush current and internal fault conditions. In Figures 7-9, UNI-QB refers to the unidirectional index, and HAR denotes the second harmonic blocking scheme. The EKF-based approach is applied only when internal faults are present. The following subsections evaluate the performance of the proposed approach under various conditions and demonstrate how

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the proposed approach addresses the problems associated with transformer differential protection schemes.

1) CT Saturation

This subsection evaluates the performance of the Com-DNN during external faults, with and without CT saturation. Figure 6 presents the primary current $i_{CT,\ target}$, and the secondary current $i_{CT,\ input}$ of both CTs, and the estimated primary current $i_{CT,\ estimate}$ during phase A-G external faults. The Com-DNN accurately reproduced the undistorted current during saturation, making this estimation suitable for protection decisions. In the following cases, the $i_{CT,\ estimate}$ is used by the Dis-DNN instead of the $i_{CT,\ target}$.

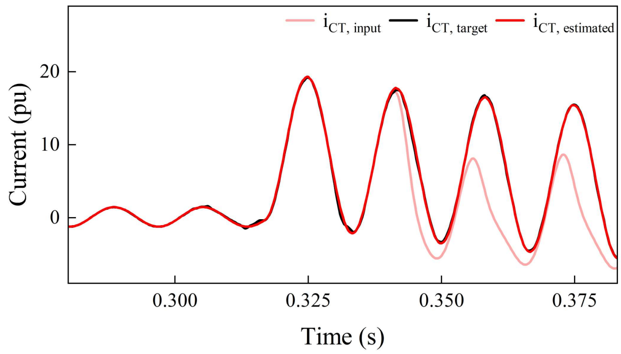

FIGURE 6. Results of CT saturation compensation using the Com-DNN

An external fault may cause both CTs to experience saturation if it is not carefully considered. Figure 7 illustrates CT saturation occurring on both CT1 and CT2 at 0.32s with a fault inception angle of 0°. Although CT2 is highly distorted and experiences saturation earlier than CT1, the actual and estimated currents almost perfectly overlap before and after the fault inception. Saturation on CT1 occurs later than on CT2, where the Com-DNN still produces promising results. The estimated currents on both CTs exhibit minor oscillations; however, these have no impact on the operation of the differential protection due to the small differential current.

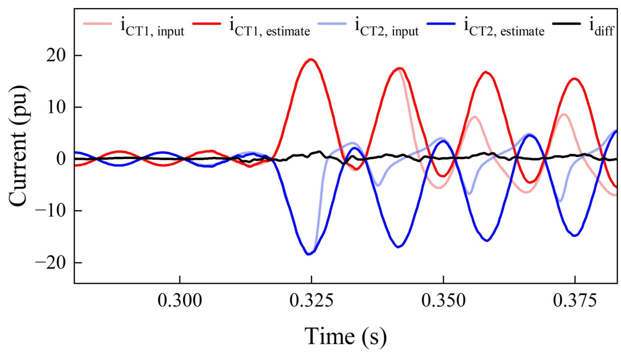

FIGURE 7. Results of the Com-DNN during an external fault on both CT1 and CT2

2) Inrush Currents and Internal Faults

This subsection describes the investigation of the performance of the Dis-DNN during inrush currents, internal faults, and the simultaneous occurrence of both. Traditional differential protection typically detects internal faults and inrush currents using harmonic-based blocking methods and dwelling-time-based methods, which are prone to errors when influenced by external factors.

Figure 8 presents the inrush current at a switching angle of 0^0 with maximum remnant flux (80%), which significantly affects the magnitude of the inrush current and produces the largest positive inrush current. HAR detects the onset of inrush current at the initial point due to the high ratio of the second harmonic. Considering the time delay, both UNI-QB and the proposed Dis-DNN respond later than HAR. UNI-QB correctly detects the inrush current with a one-cycle delay (64 samples). The Dis-DNN, with a delay of 58 samples, demonstrates a faster response than UNI-QB. It identifies the increase in differential current caused by inrush current as an abnormality captured in the differential current.

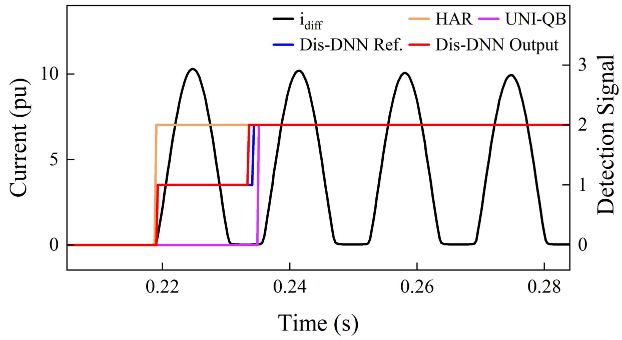

FIGURE 8. Results of inrush current detection at a switching angle of 0°

Figure 9 presents the internal fault detection at a fault inception angle of 0° and a remnant flux of 80%, corresponding to deep CT saturation. In this case, EKF exhibits the poorest performance, as it introduces oscillations during current estimation. Due to the uncertainty in threshold determination, EKF erroneously detects internal faults, resulting in false ON/OFF transitions. UNI-QB detects internal faults with a delay of one cycle. It utilizes the sum of differential currents over a window size to identify inrush current, which may result in additional detection time. Similarly, the Dis-DNN also exhibits a time delay in detecting internal faults; however, it responds faster than UNI-QB, detecting the internal fault with a delay of 58 samples. This improved speed is attributed to the Dis-DNN's ability to effectively recognize internal fault patterns. Overall, the Dis-DNN enhances internal fault detection in terms of both speed and accuracy compared to UNI-QB and EKF.

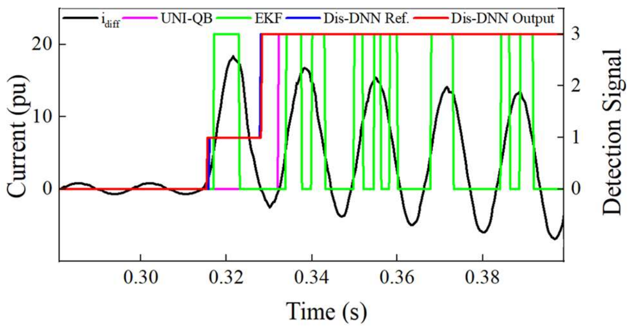

FIGURE 9. Results of internal fault detection at a fault inception angle of 0° and a remnant flux of 80%

3) *Transformer energization in the presence of internal faults*

Energizing a power transformer in the presence of an internal fault is challenging for conventional protection schemes, as harmonic-based approaches may block differential protection and potentially lead to severe damage to the transformer. Figure 10 illustrates the detection of an internal fault detection in the presence of inrush current at a switching angle of 5°. In this case, the magnitude of the fault current is greater than that of the fault current without inrush, as shown in Figure 9.

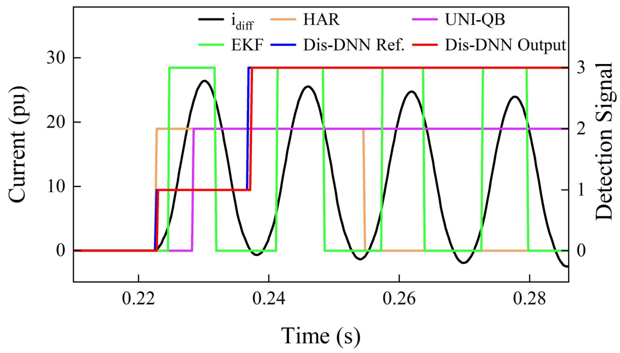

FIGURE 10. Results of internal fault detection in the presence of inrush current at a switching angle of 5°

HAR detects the second harmonic content caused by a large decaying DC offset; as a result, it mistakenly identifies the condition as inrush current for several cycles instead of correctly detecting an internal fault. Although UNI-QB identifies inrush currents and internal faults within a one-cycle delay, its performance degrades when both abnormalities occur simultaneously. UNI-QB operates based on a unidirectional index, calculated by dividing the absolute sum of the differential current by the sum of the absolute values of the differential current. This index is designed to distinguish internal faults from inrush currents. However, due to the large decaying DC offset present in the internal fault current during the first few cycles, the unidirectional index causes UNI-QB to misclassify the event as an inrush current, as shown in Figure 10. Similarly, EKF performs poorly in detecting the internal fault, producing misdetections due to noise-induced errors in current estimation. In contrast, the Dis-DNN accurately recognizes the internal fault as soon as the fault current enters its negative region, regardless of uncertain thresholds or harmonic information.

4) *Levels of Noise*

This subsection validates the performance of dual-DNN approach under noisy conditions by introducing additive white Gaussian noise at different SNR levels. Representative results are illustrated in Figure 11 for 20 dB SNR and Figure 12 for 30 dB SNR. The proposed dual-DNN approach demonstrated strong robustness against noise, with only minor deviations in detection timing. Specifically, transient states showed slightly earlier detection due to noise in the differential current, while inrush and internal fault detections were occasionally shifted by a small margin depending on the noise level. These results confirm that the dual-DNN approach is not significantly influenced by noise interference.

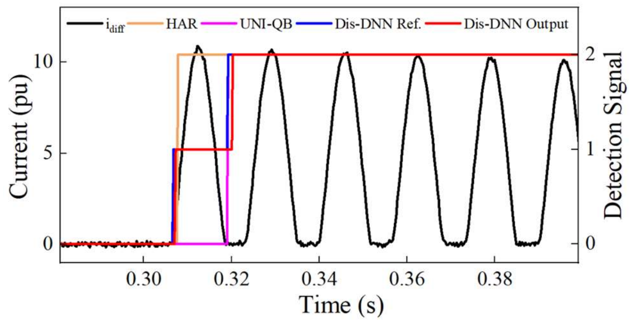

FIGURE 11. Results of inrush current detection at 20 dB SNR

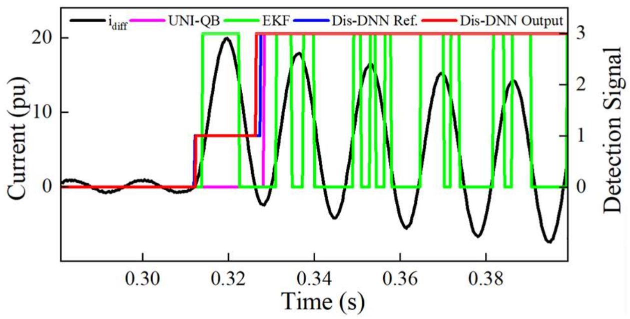

FIGURE 12. Results of internal fault detection at 30 dB SNR

5) *Transformer connection*

To validate the performance of the dual-DNN approach under different transformer connections, a delta–wye transformer was selected as a case study. Additional training is not required for transformer connections. However, to avoid potential maloperation caused by connection effects, zero-sequence removal and phase-shift compensation must be performed before applying the Dis-DNN. Following the method in [34], zero-sequence removal and phase compensation were applied to obtain the instantaneous differential currents. As shown in Figures 13 and 14, the results demonstrate that the dual-DNN approach is effective in discriminating internal faults across different transformer connections. The dual-DNN can still perform well regardless of transformer connections.

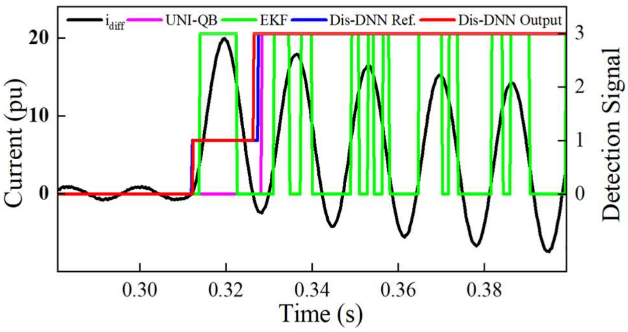

FIGURE 13. Results for a delta–wye transformer under internal fault

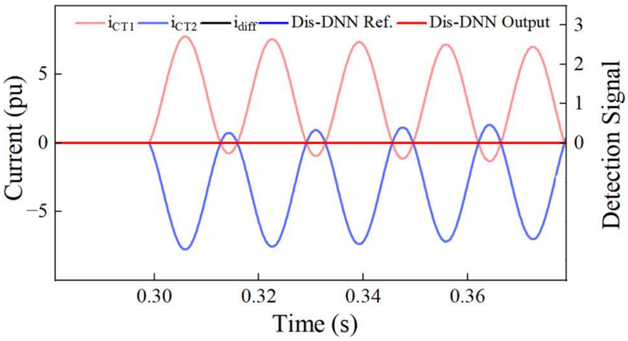

FIGURE 14. Results for a delta–wye transformer under external fault

6) *Evaluation Results*

Figure 15 illustrates the t-SNE scatter plot showing the clustering of four states and the relationships among their groups, which demonstrates how well the proposed Dis-DNN separates each class distinctively. Although some overlap is observed between the inrush and transient states during the early and late detection periods, the transient state remains generally well isolated from the normal, inrush, and internal fault groups. The normal state appears scattered mainly across the negative plane, reflecting the inherent variability in load current characteristics. Importantly, no misclassification is observed between internal faults and inrush currents.

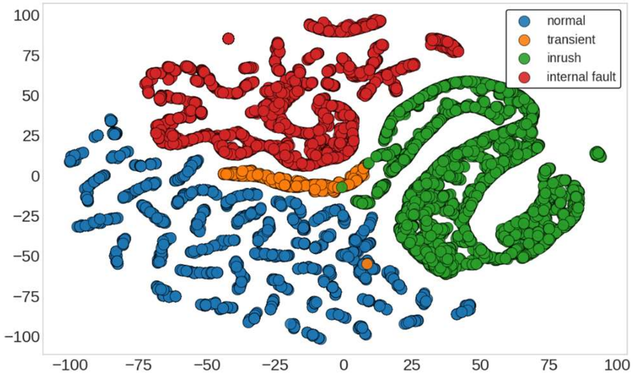

FIGURE 15. t-SNE visualization of input features in the dual-DNN approach

To further investigate the performance of the Dis-DNN, three evaluation indices (ACC, DEP, and SEC) defined in the previous subsection are used to show a numerical demonstration, as summarized in TABLE III and IV. For inrush currents, HAR demonstrates the best performance, achieving the highest values at both the 58th and 61st samples. UNI-QB also performs well in detecting inrush currents, with ACC, DEP, and SEC values of 98.12%, 93.46%, and 94.62%, respectively, at the N-6 (58th) sample. However, it does not achieve the highest metric at the 61st sample, as it sometimes detects inrush current at the 62nd sample. UNI-QB is also effective in detecting internal faults, as the presence of a negative region is generally sufficient for discrimination. Nevertheless, its performance degrades in the presence of a large decaying DC offset in the fault current.

In contrast, the Dis-DNN accurately detects inrush currents at the 58th sample, achieving the highest ACC, DEP, and SEC values of 98.87%, 99.49%, and 99.25%, respectively. At the 61st sample, it perfectly distinguishes between internal faults and inrush currents, achieving

TABLE III.
Evaluation Indices of all four methods used to discriminate inrush currents from internal faults at the N-6 (58th) sample

	Inrush current				Internal fault				Inrush current and Internal fault			
	HAR	UNI-QB	EKF	Dis-DNN	HAR	UNI-QB	EKF	Dis-DNN	HAR	UNI-QB	EKF	Dis-DNN
Accuracy (%)	100	98.12	-	98.87	-	81.26	89.18	98.75	-	42.63	90.07	99.37
Dependability (%)	100	93.46	-	99.49	-	79.52	70.61	99.62	-	41.28	72.65	99.74
Security (%)	100	94.62	-	99.25	-	83.14	75.28	98.77	-	35.29	74.91	99.01

TABLE IV.
Evaluation Indices of all four methods used to discriminate inrush currents from internal faults at the N-3 (61st) sample

	Inrush current				Internal fault				Inrush current and Internal fault			
	HAR	UNI-QB	EKF	Dis-DNN	HAR	UNI-QB	EKF	Dis-DNN	HAR	UNI-QB	EKF	Dis-DNN
Accuracy (%)	100	98.12	-	100	-	91.08	93.84	100	-	45.12	92.38	100
Dependability (%)	100	96.14	-	100	-	89.43	73.56	100	-	48.47	74.63	100
Security (%)	100	98.71	-	100	-	87.69	78.43	100	-	50.15	85.63	100

100% in all three metrics. The Dis-DNN also performs well under inrush conditions involving internal faults. At sample N-6, it outperforms the other three methods, with ACC, DEP, and SEC values of 99.37%, 99.74%, and 99.01%, respectively. At sample N-3, the Dis-DNN again achieves perfect performance, with all three metrics reaching 100%.

Compared to the Dis-DNN, EKF performs poorly. It misclassifies internal faults due to discrepancies between the reference and estimated currents. Moreover, its heavy reliance on threshold settings limits its general applicability. As a result, EKF struggles to distinguish between inrush currents and internal faults. At sample N-3, its ACC, DEP, and SEC values drop to 93.84%, 73.56%, and 78.43%, respectively. As previously mentioned, its performance degrades particularly when the internal fault current contains a large decaying DC offset. In such cases, EKF may block the differential protection for several cycles until sufficient negative regions appear in the waveform.

IV. Hardware Implementation

This section describes the hardware testing with an implementation based on the TMDSIDK574 (AM574x Industrial Development Kit), which includes two C66x DSP cores and two ARM Cortex-A15 cores. For signal sampling, the TMDSIDK574 is interfaced with the AD8688EVM via SPI. The AD8688EVM operates at a sampling rate of 64 samples per cycle and features a 16-bit resolution analog-to-digital converter. The testing configuration is illustrated in Figure 16.

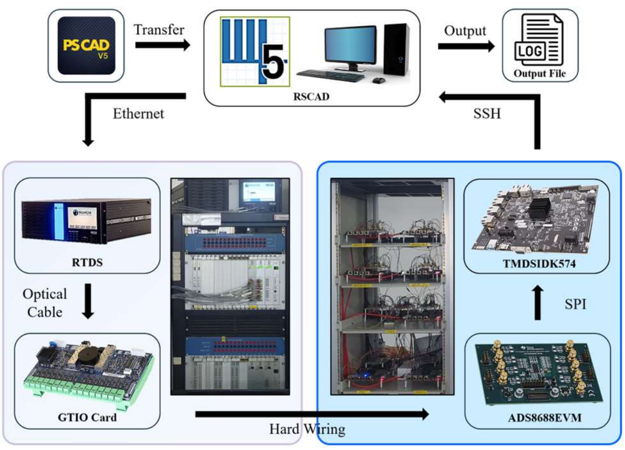

FIGURE 16. Testing configuration for hardware implementation

To implement and evaluate the proposed dual-DNN approach on the TMDSIDK574, the model is reconstructed based on its mathematical formulation and integrated into a custom implementation that processes real-time ADC input data for inference on a single C66x DSP core. Each DSP core has a computational capability of approximately 12×10^9 floating-point operations per second (FLOPS) and a clock frequency of up to 750MHz. Given that the dual-DNN approach requires approximately 28,353 floating-point operations (FLOPs) per inference, the estimated inference time on a single DSP core is approximately 2.36µs. To enhance computational efficiency, the C66x DSPLIB is employed with cache optimization and loop unrolling techniques, which reduce computational overhead and improve execution speed. Furthermore, the model precision is reduced from FP32 (32-bit floating point) to INT8 (8-bit integer), thereby minimizing memory usage and increasing inference throughput.

Figure 17 and 18 compare the simulation and hardware testing results when the Dis-DNN identified an A-G fault and inrush current. Compared to the simulation results, the detection of both internal faults and inrush currents was slightly delayed due to hardware-related factors such as ADC errors and quantization noise introduced by the Com-DNN, which caused slight oscillations. These factors can lead to inaccurate signal amplitude representation and waveform distortion, resulting in delayed fault detection. Despite these challenges, the proposed approach remained stable and successfully detected both internal faults and inrush currents. Although the Com-DNN introduced minor oscillations in the differential current, they had no significantly detrimental effect on the performance of the Dis-DNN.

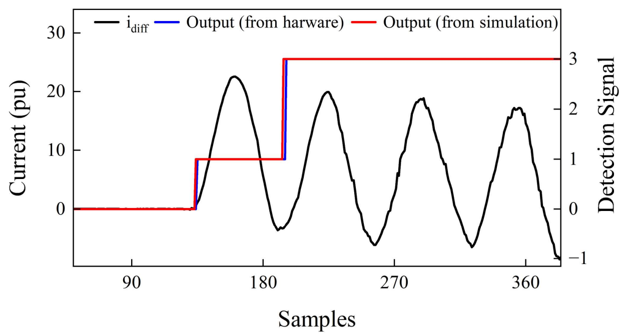

FIGURE 17. Results of Internal fault detection in simulation and hardware testing

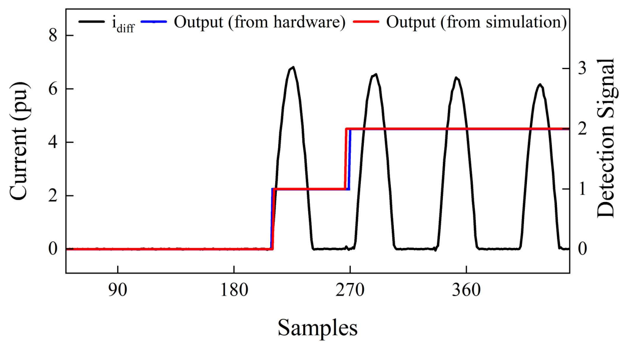

FIGURE 18. Results of Inrush current detection in simulation and hardware testing

V. Conclusion

In this paper, we proposed a dual-DNN approach, comprising a Com-DNN and a Dis-DNN, to reliably discriminate internal faults from simultaneous inrush currents in transformer differential protection, even under CT saturation conditions. The Com-DNN effectively mitigates the adverse effects of CT saturation by recognizing and compensating for distortions in current signals, thereby enhancing dependability and security prior to the discrimination phase. The Dis-DNN subsequently distinguishes between inrush currents and internal faults with high accuracy under the different level of noise and transformer connection.

The performance of the proposed approach was evaluated on a 23-kV power transformer under various scenarios, including simultaneous inrush and fault events. Simulation results demonstrated robust dependability and security compared to a conventional ANN and three other existing methods. The proposed approach exhibited no erroneous activation of differential protection, and its operational time delay remained within an acceptable range for practical power system protection. Furthermore, hyperparameter optimization using BO enhanced the generalization capability of the proposed approach.

Hardware implementation on a TMDSIDK574 IDK validated the feasibility and effectiveness of the dual-DNN approach, confirming its potential for real-world transformer protection applications.

The main limitation of the dual-DNN approach lies in its adaptability to power transformers with distinctive B-H characteristics. In addition, the handling of sympathetic inrush currents remain a challenge for this proposed dual-DNN, and it will be addressed in future work and will be adapted to maintain robustness under such conditions.

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